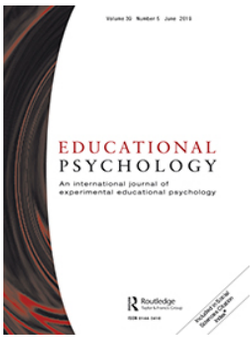

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Differences of math anxiety groups based on two measurements, MASS and EEG

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ABSTRACT

This article investigated how mathematic anxiety (MA) of Korean middle school students could be reduced by comparing analytically their cognitive neuroscience and questionnaire results. We developed a three-hour Complex Treatment Program (CTP) on quadratic functions for the study. In the summer of 2016, we collected data of the pre and post MA questionnaires by Mathematics Anxiety Scale for Students (MASS), the percent of correct answers (PCA) and reaction time (RT) by E-prime program, and also brain-imaging data of the event related potentials (ERP) by Electroencephalograph (EEG) using computer-based functional F-G model. The result indicated the CTP to be effective with the group with higher math anxiety and the group with higher achievement respectively. The MASS result was verified with the better performance of PCA on type G, which was measured by E-prime program. Some interesting patterns were revealed on brain-imaging data by EEG, indicating more brain activities with the MA groups.

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Mathematics Anxiety (MA); Mathematics Anxiety Scale for Students (MASS); Mathematics Achievement; Functional Task; Electroencephalograph (EEG); Complex Treatment Program (CTP)

Introduction

In Korea, a student who loses interest in mathematics has been referred to as a 'supoza' and the rising number of supoza has been a growing social concern in the country for the past decade. The non-governmental organization called 'NOWORRY21' (2015) reported that a survey of 7719 students resulted in 36.5% (6th grade), 46.2% (8th grade), and 59.7% (12th grade) of supozas in school. Because mathematics is a fundamental subject in primary and secondary school, becoming a supoza at the middle-school ages can have a negative impact on other classes in school. If this learning experience influences the formation of the self-concept such as self-confidence or self-esteem which is related to mathematical learning attitude (Gourgey, 1982), these accumulated experiences could negatively impact students' attitudes towards the learning of mathematics.

Looking back at the Programme for International Student Assessment (PISA) from 1995 to 2012, Korean students ranked in the highest level of achievement in

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Table 1. Definitions about mathematics anxiety.

Researchers	Definitions of mathematics anxiety
Dreger & Aiken (1957)	Emotional signs on arithmetic or mathematics
Richardson & Suinn (1972)	Tension and anxious feelings that interfere with problem solving in everyday life and learning situations
Tobias (1976)	When solving mathematical problems, situations such as fear of the individual, mental anxiety
Fennema & Sherman (1976)	Anxiety and fear related to math learning
Byrd (1983)	Situations in which an individual feels anxious when they are in any way mathematical situations
Uusimaki & Kidman (2004)	Fear of not having the knowledge or expertise to be successful in completing a mathematical task regardless of whether is simple or complex
Sheffield & Hunt (2006)	Feelings of anxiety that some individuals experience when faced with mathematical problems

mathematics but consistently marked a very low score in affective domains like self-efficacy and interest (Ko & Yi, 2012). Since 2012, they have not shown any improvement in this area of national concern. This leads us to conclude that students who are high-performers in mathematics could potentially have low achievement levels in the affective domains of mathematical learning, resulting in mathematics anxiety (MA) which is displayed as the uneasy feeling about the subject of mathematics.

While many previous studies about MA were centered on the development of questionnaires to diagnose students' MA, recent studies have been focused on devising a treatment to abate it. For example, researchers applied the Cooperative Learning (or Groups) Model (Daneshamooz & Alamolhodaei, 2012; Won, 2009), the Note-taking Model like Divided-page Method (Eun & Lee, 1994; Tobias, 1987), Cornell-note Method (Jung & Whang, 2013), and the Mathematics-friendly Program (Yoon & Jeon, 2010) to students who were feeling mathematics anxiety. However, these studies focused mainly on problem solving or simple algorithms and not on a specific domain of mathematics, e.g. function. In order to reduce MA in the students' functional understanding of mathematics, we developed a complex treatment program that combined a mathematical treatment and a psychological treatment, once used by Yoon and Jeon (2010). Moreover, we collected data to grasp the effects of the program more closely by measuring the students' brain waves which is the difference of this study from previous ones.

The purpose of this study was to investigate how to reduce the MA of eighth-grade students after they completed the complex treatment program (CTP), which was a mixture of psychological and mathematical treatments to help students psychologically and mathematically. To find the results, the research was carried out by comparing the differences among the groups based on the Mathematics Anxiety Scale for Students (MASS) by Ko and Yi (2011) and on analyses of data through E-prime program that randomly chooses an item and presents it to a student on the computer monitor, and the event related potentials (ERP) of the electroencephalograph (EEG). The research questions were set up as follows:

1. How did the student groups respond to the MASS before and after the complex treatment program?

2. What were the differences of the groups in the percent of correct answers (PCA) and reaction time (RT) gained by E-prime program while getting the brain imaging data through ERP?
3. What were the differences in the brain imaging data of the groups on the functional tasks through ERP?

Theoretical background

Mathematics anxiety (MA)

In the previous research, various researchers developed differing definitions of MA which are summarized as follows.

Since Byrd's (1983) comprehensive definition, several researchers have defined MA differently and others who were not in Table 1 mostly used the definition formerly described in the literature. Various definitions, however, were similar in the context of mathematical learning. In this study, we defined MA as "anxiety felt by individuals when they solve mathematical problems".

As well as defining MA in many ways, the sub-factors characterizing MA vary from one researcher to another. Huh (1996) analyzed the factors of MA presented by various researchers and revised them to five main factors: 'math as a subject', 'math achievement', 'cognitive & negative thought', 'attitude toward math', and 'teacher' with 106 items. More recently, Ko and Yi (2011) modernized and simplified the previous MASS including Huh's into 65 items with four main factors and 12 sub factors. This questionnaire was used because it was suitable for middle school students.

Hembree (1990) presented an integrated result by meta-analysis of 151 previous mathematics anxiety studies. The result of his research listed the systematic desensitization of behavioral treatment methods, cognitive modification of self-confidence, adjustment of cognitive beliefs, cognitive restructuring, combined systematic desensitization, and relaxation training. These were summarized into the psychological treatment, non-psychological treatment, and the complex treatment which was a mixture of psychological and non-psychological treatment presented later by Huh (1996).

Many studies focused on reducing MA with the effect of the treatment based on various teaching and learning methods. Daneshamooz and Alamolhodaei (2012) investigated how much cooperative learning could be helpful for college students with high math anxiety, compared to the traditional learning. Eun and Lee (1994) reported that the students using the note-taking on divided-pages method suggested by Tobias (1987) became better solvers of word problems due to the effect of reducing MA. Jung and Whang (2013) analyzed both Cornell-note-method and Divided-note-method, and found they were effective in reducing MA in problem solving. Yoon and Jeon (2010) developed a math-friendly activity program geared toward easing MA and checked the effect of the developed program on the alleviation of mathematics anxiety of 4th grade students, which consisted of psychological treatment (positive thinking training) and non-psychological treatment (mathematics activities). As a result, the collective averages of students on MA and its sub-factors were lower in the posttest than in the pretest.

The researchers in the previously mentioned literature developed treatments using various teaching and learning methods to reduce students' MA and reported that most of them were effective to achieve their purpose, but they mainly dealt with 'mathematical' problem solving or simple algorithms and not with a specific area of understanding such as functional understanding. Also, as shown by Yoon and Jeon (2010) the research using a complex treatment which is a mixture of psychological and mathematical treatments has not much been carried out yet in the field of MA.

Brain research on students' understanding of functions

In this study, we focus on mathematics anxiety in the context of solving mathematical tasks related to functions, which play an important role in the development of algebraic reasoning. Reasoning with functions is one of the core strands within algebraic reasoning as identified by Kaput (2008) and National Council of Teachers of Mathematics (2000). Akkoç and Tall (2002) suggested the complexity of the function concept arises from its many representations, e.g. graphs, equations, and tables.

In our study, we decided to focus on the translation between graph and equation representations—a particularly important concept in understanding functions (Williams, 1998). As such, our study resembles two previous neuroscience studies on students' understanding of functions (Thomas, Wilson, Corballis, Lim, & Yoon, 2010; Waisman, Leikin, Shaul, & Leikin, 2014) though neither of them considered the effects of MA.

Thomas et al. (2010) classified function tasks into four formats: graph-to-graph, equation-to-equation, graph-to-equation and equation-to-graph. They used these formats for both linear and quadratic functions, generating a total of eight kinds of tasks for the ten students in their study—college students in New Zealand. In addition to measuring RT and PCA, they measured neural activity within the neocortex of the brain through use of functional magnetic resonance imaging (fMRI). Essentially, fMRI measures neural activity by way of blood flow to areas within the brain.

They found a significant increase in reaction time and a significant decrease in accuracy for the two cross-representation formats (graph-to-equation and equation-to-graph) when compared to the two same-representation formats. However, there were no such differences when comparing the two cross-representation formats nor when comparing the two same-representation formats. Surprisingly, participants responded more accurately to tasks involving quadratic equations than to tasks involving linear equations. In a follow-up interview, all participants reported that, in general, they solved tasks in the cross-representation format by focusing on key features, such as intercept and slope.

Regarding neural activity, Thomas et al. (2010) found parietal areas previously associated with number and arithmetic (the intraparietal sulcus [IPS] and the posterior superior parietal lobule [PSPL]) involved in processing of mathematical functions in both graphical and algebraic formats. These brain regions were even more active during cross-representations format tasks. The researchers concluded that the overall neural activity during the tasks suggests involvement of spatial and verbal working memory in the translation of functions.

Waisman et al. (2014) conducted a similar study with high school students in Israel. In their research, the students were divided into four groups: students identified as gifted who excelled in mathematics (G-EM), students identified as gifted who did not excel as mathematics (G-NEM), students not identified as gifted who excelled at mathematics (NG-EM), and students not identified as gifted who did not excel at mathematics (NG-NEM). These researchers used EEG technology to analyze peaks in ERPs. The functional tasks were similar to those used by Thomas et al. (2010). From these results, the researchers conjectured that, "problem-solving expertise developed by students without general gifted-ness is achieved by means of high cognitive effort" (p. 690).

In summary, few studies in the area of brain research have found the characteristic of MA which was the main concern in our study. We wanted to know what might take place in brain while students with MA were working with functional tasks after a complex treatment program was implemented. This program for the study would thoughtfully be developed to help students with MA so that they become confident in doing mathematics.

Method

Participants

Initially, thirty students in the eighth-grade were chosen based on teacher recommendations from a middle school in Yongin, a small town located thirty miles south of Seoul, Korea. Most of their parents belonged to the middle class of the Korean society. According to the schedule of the middle school curriculum supervised by the local ministry of education, these students finished learning quadratic functions a month before the research began. The students had never participated in a program for the purpose of reducing MA and they requested the research results be provided to them on completion of the experiment. They turned in the consent form with the approval of parents.

Unfortunately, five students were excluded in the final data analysis. Three students' brain waves using EEG was not successful due to poor reception to the receiver worn around the subject's head and two students were absent on the day of the post-questionnaire. Therefore, a total of 25 students' data was collected in the research.

Procedure

To investigate the effectiveness of the treatment program in this study, we used the pre and post questionnaires for MA and then collected PCAs, RTs and brain-imaging data. The data were collected during July in 2016 on which the subjects were available right before the summer vacation started in Korea. In this study, the pre and post-questionnaires by MASS were used to measure students' MA. In the same way as Sheffield and Hunt (2006), the participants were classified as high MA (HAX) group or low MA (LAX) group based on the result of the pre-questionnaire. The mean value of anxiety on it for 25 students was 2.63. The students who were higher or lower than 2.63, were classified as either HAX or LAX respectively. Also, students who were higher or lower than the mean value of the final exam administered by their school, were

Table 2. Distribution of the groups.

	HAC	LAC	Total
HAX	4	7	11
LAX	11	3	14
Total	15	10	25

grouped as either high AC (HAC) group or low AC (LAC) group to see relationship between cognitive and affective aspects in detail. The distribution of these groups is shown in Table 2.

The treatment program of the three-lesson units was applied to these students. The students reviewed the problems on their own, while working with the experimenter. To find the effectiveness of the treatment program, we investigated the degree of MA using the same questionnaire as the pre-questionnaire and measured the students' brain waves to see how they reacted to stimuli of functional tasks while they solved them on the monitor provided.

Instruments

Questionnaire for MA

The Mathematics Anxiety Scale for Students (MASS) revised by Ko and Yi (2011) was used to measure students' MA. This test scale was chosen because it was more appropriate to the middle school students since their MASS was developed for the secondary school students including middle school students with 65 question items which resulted in the internal consistency reliability, the Cronbach's α = .976 as reported. Each question was answered on one of 1 to 5-point Likert scale.

Complex treatment program (CTP)

The complex treatment program in three units consisted of the mixture of a mathematical treatment and a psychological treatment. The mathematical treatment that was composed of four kinds of problems (see Appendix 1) provided students a chance to reflect the nature of quadratic function. Firstly, the diagnosis problem was constructed as the basic concepts learned in the previous lesson. Secondly, a lower level (basic) problem was constructed with a basic concept of the quadratic function. Thirdly, the treatment was constructed to help students to understand the type G. Lastly, a higher level (challenging) problem was constructed to challenge advanced level students with a higher level of difficulty. Also, four to six mathematical-hint cards reflecting Vygotsky's scaffolding theory in which they could be assisted by peers or a teacher whenever needed were prepared for each problem (see Appendix 2).

The psychological treatment which was the other part of CTP was constructed by revising the brain integration in education (BIE) program used by Kim (2010) to reduce the students' anxiety level shown in the pre-questionnaire. Kim (2010) developed the brain integration in education (BIE) program consisting of three stages: understanding of brain, integration of brain, and application of brain, to improve self-directed learning ability of middle school students. The findings suggested that the BIE program implemented for eight weeks to help use both right and left brains in an integrative, balanced way could stimulate brain development of students. We developed the part

Table 3. Complex treatment program.

Unit	Treatment	Contents	Brain/Mathematics Education
	Pretest	• Perform mathematics anxiety pretest	
1	Introduction	• Introduce a complex treatment program	
	Psychological	• Think about the need for learning math in relation to future jobs using a clip.	Understanding of Brain (Brain's Potential/Worth)
	Mathematical	• Suggest problems with guidance • Use hints to solve problems	$y = ax^2$ $y = ax^2 + q$
2	Psychological	• Find a clue or clues for handling any issue by asking a peer to help each other.	Integration of Brain (Main Agent of Information)
	Mathematical	• Suggest problems with guidance • Use hints to solve problems	$y = a(x-p)^2$ $y = a(x-p)^2 + q$
3	Psychological	• Act some skit to improve your patience.	Application of Brain (Brain's Practice)
	Mathematical	• Suggest problems with guidance • Use hints to solve problems	$y = ax^2 + bx + c$
	Posttest	• Perform mathematics anxiety posttest	

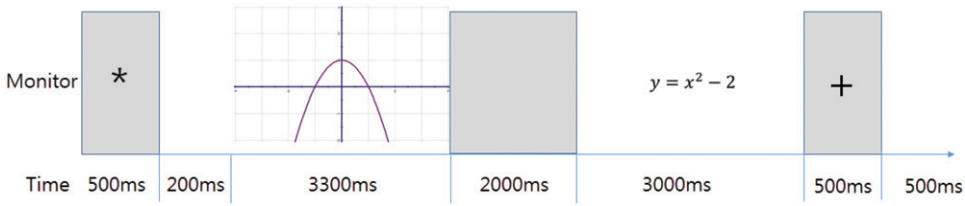

Figure 1.. One Sample of Type G.

of psychological treatment of the CTP based on Kim's BIE, to activate brain education in function (see Table 3).

F-G model of functional translation.

Thomas et al. (2010) presented the functional tasks in relation to graphs and equations as stimuli and observed 10 students' brain waves. By actively using it, the tasks developed in this study were composed of 10 kinds of basic quadratic functions with $a = \{-1, 1\}$, $p = 0$, $q = \{-2, -1, 0, 1, 2\}$, in an algebraic formula, $y = a(x-p)^2 + q$. In other words, the subject was able to grasp the elements such as shape, intercept, vertex, etc. of a quadratic function.

Two hundred items were developed which asked students to decide whether graphs and equations matched. E-prime program randomly selected an item and presented it to the subject. Two types of cross-format functional tasks were translating from equation-to-graph (type F) and translating from graph-to-equation (type G) using the ten quadratics functions. The students solved 20 questions as presented randomly for each type. Figure 1 below shows a timetable for type G as an example. In the F-G model, as soon as they solved 20 type F problems, the student was ready to solve type G. The test followed a set, timed sequence. First, the student pressed the spacebar. An '*' appears on the monitor for 0.5 s, which was an indication that the item would start. Then, it displayed a blank screen for 0.2 s, a graph was presented for 3.3 s, and then the screen went blank for 2 s. Subsequently, it presented an equation for 3 s. At this time, the user was to decide whether the graph and algebraic expression

matched. The '+' sign also appears for 0.5 s to indicate the next item. Type F questions which preceded type G, were delivered in the same manner except the graph and the equation sequence was reversed.

All students started by solving the type A arithmetical tasks (20), then type F (20), and finally type G (20), which took them a total of 10 min. This procedure was an 'F-G Model'. Doing a mathematical task, compared to doing a task in the other disciplines takes a higher degree of cognitive understanding (Skemp, 1987). For the beginning, 20 arithmetical tasks (defined type A) were designed to activate the brain as a warm-up stage for the appropriate measurement of the functional tasks as used in Sheffield and Hunt (2006). By contrast, some research like Thomas et al. (2010) did not consider the role of a warm-up stage in their study. As a result, their findings as reported by their PCA and RT gained from the translation activity of either linear or quadratic functions were contrary to the findings of previous studies (e. g., Duval, 2006): In general, we might expect translation of quadratic functions to be more difficult than linear ones mathematically. Regardless of its role, we did not include the comparison of arithmetic with functional understanding in the study, but only focused on functional understanding at this time.

Brain-imaging technique.

For this study, Electroencephalograph (EEG) was selected. This decision was based on budget, mobility, accessibility, safety, functionality, and practicality. In detail, we selected ERP among the EEG's techniques. ERP is a method of averaging EEG activity from time domain analysis and multiple-stimulation in an EEG. It usually appears within 50–500 ms after stimulation: P50, and N100 which reflect the stimulus detection, and cognitive-related factors, P300, and N400 which appear later and mainly indicate stimulation.

Results

Students' responses to MASS

Due to the small sample size, group differences on the questionnaire, MASS were tested using a non-parametric test (Wilcoxon ranks). As the result of the pre-questionnaire, 11 out of 15 students with HAC level reported LAX while four participants displayed HAX. On the other hand, of the ten LAC students, seven had HAX with only three in the LAX group (see Table 2). This suggests that students with higher achievement scores feel less anxious about mathematics. In fact, the results of Wilcoxon rank-sum test for two independent samples showed us the comparison of degree of MA in terms of achievement as follows: HAC's mean rank of MA was 9.83 ($SR = 147.5$), while LAC's mean rank of MA was 17.75 ($SR = 177.5$). *SR* stands for sum of ranks. There was a significant probability, ($z = -.2635$, $p = .008$) in the degree of MA according to the achievement levels of HAC and LAC at the significance level .05.

Wilcoxon matched-rank test was used to see the effect of the complex treatment program in which two matched samples were drawn either from identical populations or from symmetric populations with the same mean (Howell, 2002). This can occur when we wish to investigate any change in scores from one time point to another, or

Table 4. Wilcoxon's signed-ranks test.

Group(post-pre)	N	Z	Asymp. Sign.
MA			
HAX	11	−2.193(a)	0.028*
LAX	14	−1.162(a)	0.245
AC			
HAC	15	−2.158(a)	0.031*
LAC	10	−1.007(a)	0.314
Total	25	−2.258(a)	0.024*

Note. (a) stands for "Based on positive ranks."

when individuals are subjected to more than one condition. The following Table 4 compares the results of the pre-questionnaire by groups with the results of the post-questionnaire after the program was implemented.

The CTP was effective since z value was −2.258 with significance probability, 0.024 ($p < .05$) for the total of 25 students. By groups, z value was −2.193 with 0.028 (p-value) for the HAX students and was −1.162 with 0.245 (p-value) for the LAX students. Also, z value was −2.158 with 0.031 (p-value) for the HAC students and was −1.007 with 0.314 (p-value) for the LAC students. In summary, the groups of HAX and HAC were significantly helped by the CTP. In contrast, the differences of the pre and post-questionnaires of the groups, LAX and LAC were not statistically significant even if their anxiety levels were lessened.

The percent of correct answer (PCA) and reaction time (RT)

In order to measure the responses of the students who participated in the research, we developed the functional types for the E-prime program along with EEG measurement, which provided us the percent of correct answers (PCA), and the reaction time (RT) of the subjects. Before solving the F-G Model, the students solved arithmetical tasks (type A) consisting of simple addition problems to help activate their brain. This is called the dual-task composition in which the students worked on two phases: first, the phase for arithmetical tasks; second, the phase of functional tasks to capture more proper activation of brain. However, the result of type A was not included in this paper. (The authors plan to report it later in comparison to the functional tasks.) PCAs and RTs of four groups for each type of tasks are as follows (Table 5).

In the case of type F after working with CTP, the PCA and the RT of group LAX were better than those of group HAX. Similarly, the HAC group scored better than the LAC group in terms of PCA and RT for type F. Moreover, in the achievement (AC) groups, always group HAC did better performance than group LAC as we expected on both types of functional translations after CTP. However, group HAX of the anxiety (MA) groups, showed less performance than group LAX in only type F, but not in type G.

In the case of type G after working with CTP, PCAs and RTs of all groups showed better performances in comparison to type F. This result is different from the result by Han and Choi-Koh (2016) with high school students who had the psychological treatment only. In this study, we developed a complex program with a mixture of the psychological and mathematical treatments. Surprisingly, group HAX performed better

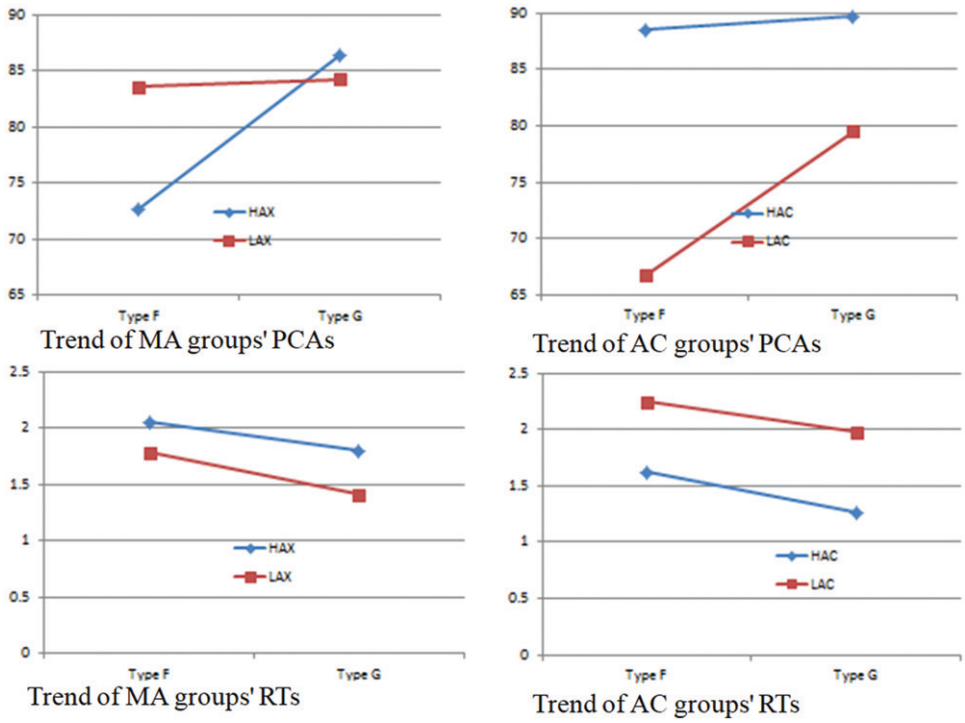

Figure 2.. The Trend of Patterns on PCAs & RTs.

than group LAX in type G where CTP provided more chances onto type G, even though the group HAX took a little more time. That is to say that the CTP helped the group with higher degree of MA to do a better job on type G in the study. In appearance, group HAX of MA groups and group HAC of AC groups indicated better results which meant that they were enhanced more than the counter parts on type G. However, what we should notice is that group LAX and group HAC did show a smaller gap in the difference between type F and type G on the upper graphs in Figure 2. That is, group LAX and group HAC were not affected by the types as much as their counterparts in the other groups.

In the graphs on the right hand in Figure 2, the AC groups showed larger gaps than the MA groups on both the PCAs and RTs of the groups for the two types, although group HAX exceeded group LAX on type G. In terms of RTs, the pattern of the distribution looks similar, showing that both AC groups and MA groups took a little longer on type F. As mentioned earlier, mathematical treatment was reinforced with the problems that provided more chances to work on type G based on the result of the former study (Han & Choi-Koh, 2016; Thomas et al., 2010). Regarding type G, the mathematical treatment played an important role of improving their performance, especially for either group HAX, or group HAC from each aspect. In summary, the finding that group HAX exceeded the counterpart in PCA on type G, meant that its degree of MA must have been lowered to produce the better result than group LAX.

EEG

In this study, we analyzed P300 wave in ERP to measure brain waves, in addition to PCAs and RTs by E-prime program. The P300 wave is a positive wave that reaches the apex about 300 ms after stimulus presentation starts in ERP. The results of measuring brain waves while solving types F and G through EEG are shown in Figure 3 which compares the groups for the types.

After having completed the planned CTP, the P300 wave of the HAX group was higher than that of the LAX group as shown in the top two graphs in Figure 3. This result indicated that the activity of the working memory used in the brain was larger with group HAX versus group LAX. In other words, group HAX was forced to expand more mental activity than group LAX, while performing the same functional tasks. Even in arithmetic, Sheffield and Hunt (2006) discovered much a higher amplitude of mental activity with the HAX group, compared to the LAX group, when carrying was involved in addition problems.

On the other hand, in the lower graphs of Figure 3, the P300 brain waves of the HAC and LAC groups had similar patterns regardless of the translation type. Based on this result, although it may be somewhat difficult to generalize due to the outcome of a small sample size, the differences affecting EEG were larger in MA groups than AC groups. Figure 4 compared the EEG by the type regarding the groups.

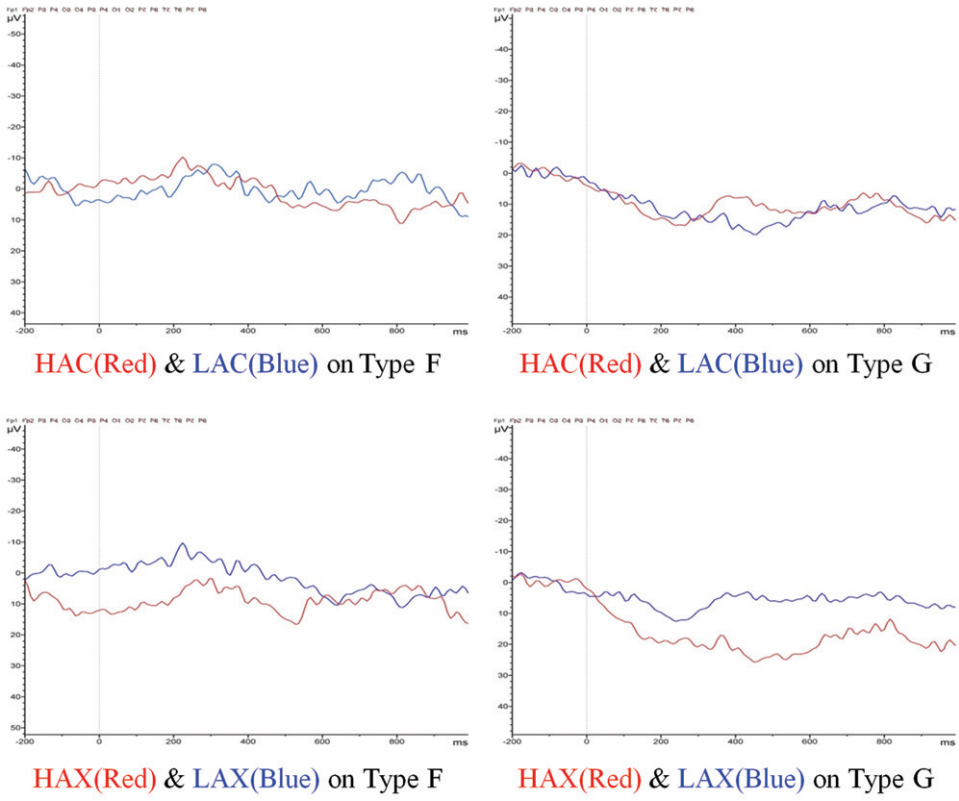

Figure 3.. The EEG by Groups for Types.

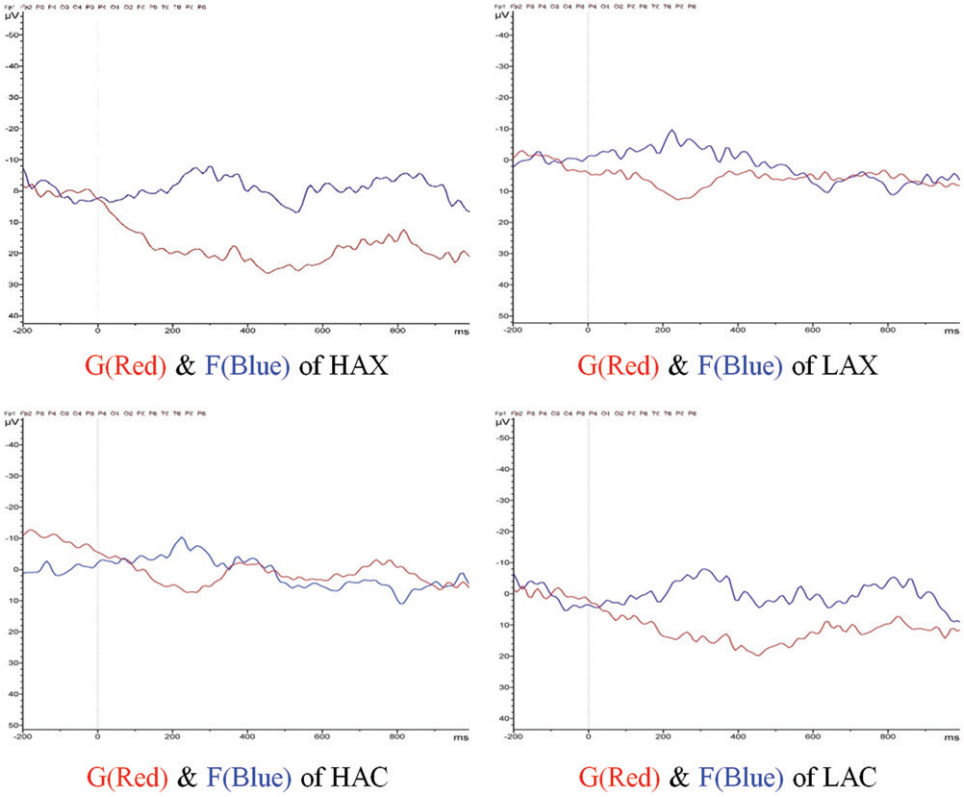

Figure 4.. The EEG by Types for the Groups.

In Figure 4, the P300 wave while solving type G was higher than the related P300 wave for type F regardless of the group. In a diagonal direction in Figure 4, there is a similarity in the pattern of the two types in groups, LAX and HAC, and in groups, HAX and LAC, which shows some connection between these groups. A lot of research showed the negative correlation between MA aspect and AC aspect (Rampersad, 2003; Zakaria, Zain, Ahmad, & Erlina, 2012; Ma, 1999). Especially, group LAC showed little improvement by the treatment as shown on the mark of the higher amplitude with the lowest score of PCA in Table 5 than the counterpart. Also, group HAX displayed the highest amplitude among the groups. So, groups LAC and HAX were connected with a higher amplitude which indicated more activity.

The PCA was relatively better and the RT was shorter on type G than type F in Table 5. However, the comparison of the brain waves showed that there seemed to be more activities in the brain, showing higher amplitude in the process of solving type G than type F. This is an example of the importance of EEG measurements. If only PCA and RT had been measured without EEG, we could have been convinced that students solved type G better than type F after working with CTP. It, however, could be seen by measuring EEG that type G made students undergo more mental activities than type F because the students had to try to decode the image of a graph first with a high degree of concentration in order to relate the information with a corresponding algebraic equation. This might be associated with the finding of Thomas

Table 5. PCA and RT of groups on types (*SE* in parentheses).

	Type F		Type G	
Group	PCA (%)	RT (sec)	PCA (%)	RT (sec)
HAX	72.7 (3.9)	2.05 (0.14)	86.4 (3.0)	1.8 (0.11)
LAX	83.6 (2.9)	1.78 (0.11)	84.2 (2.8)	1.41 (0.07)
HAC	88.5 (2.5)	1.62 (0.11)	89.7 (2.4)	1.26 (0.07)
LAC	66.7 (4.1)	2.25 (0.14)	79.5 (3.5)	1.98 (0.11)

Note. *SE* is the abbreviation of standard error.

et al. (2010) that the regions of intra-parietal sulcus (IPS) and the posterior superior parietal lobule (PSPL) in brain were rigorously activated on type G.

Conclusions and suggestions

Conclusions

In order to reduce the mathematics anxiety of the middle school students, we developed and applied the CTP that combined a mathematical treatment that aimed to strengthen functional understanding and a psychological treatment that aimed to improve psychological stability for the students. At the end, we measured students' brain waves using EEG, while they were solving given types of functional tasks. The results are summarized as follows.

First, the test results by MASS showed that our program overall was effective for the students to reduce MA. In specific, MA of group HAX and group HAC had significantly been lowered (see Table 4). In fact, because our program was composed of two important parts that help students with psychological and mathematical aspects, this outcome made sense to us. When we classified them into groups based on students' scores of MASS pre-questionnaire and of their recent regular exam in school, it indicated that 11 students out of 15 HAC group might have a sense of easiness towards the subject of mathematics with low levels of MA, while seven students out of 10 LAC group, felt uneasy towards mathematics. For the relationship between MA and AC was considered important in planning this study, the AC groups were included into the study.

Second, by analyzing PCAs and RTs by E-prime program, we discovered that the students belonging to either the HAX group or the HAC group produced better PCAs in type G and either the LAX group or the HAC group did better PCAs and RTs in type F through the CTP than their counterparts respectively, although the program was composed of three-lesson units over a short period of time. According to the previous research by Han and Choi-Koh (2016) with high school students and Thomas et al. (2010), and Seok (2015) with college students, the students solved type F more accurately and quickly than type G. Also, they reported that the students found type G more complicated than type F. Based on their results, the mathematical treatment part of the CTP was designed to strengthen type G more than type F for this study (see Appendix 1). After completing our program, all the groups of students solved type G more accurately and quickly than type F. That is, the CTP we had developed to help the students not only to reduce MA, but also to get over the difficulty on type G for better functional understanding played an important role in achieving the purpose

of the study. Interestingly, the result by E-prime that HAX and HAC exceeded the counterpart on the PCAs of type G in Table 5 seemed to verify the result we got by MASS questionnaire in Table 4.

Third, to examine the precise effect of the program, we observed closely at the brain waves by EEG while the students were solving the F-G Model. As mentioned above, although the students solved type G more quickly and accurately, compared to type F, we noticed that the students had more brain wave activity with type G in brain (see Figure 3). Also, the MA groups presented a bigger gap between two types, which meant more activities on type G in brain. If we had not measured EEG in addition to the questionnaire of MA, we might not have found this kind of difference with type G in which students were asked to decode the graph image first, and then to match it to a corresponding equation of function. Therefore, the MA aspect influences more on students' comprehension of images in performance.

In Figure 4, we see some similarity in patterns in diagonally across groups. The group HAX that was a lot helped by CTP on type G showed a similar P300 brain wave pattern with a bigger gap between the types to the group LAC that was rarely helped by CTP in this study. On the other hand, the group HAC that was a lot helped by CTP on both types showed a similar pattern to the group LAX, with even a bigger gap between two types. So, two groups, HAX and LAC and two groups, LAX and HAC, respectively, showed a similar pattern on the brain-imaging data, leading to the result of a negative correlation between MA aspect and AC aspect, which was also evidenced by Ma (1999), Zakaria et al. (2012), etc.

Suggestions

This is a pioneer study which used brain-imaging data in addition to MASS data in analysis to gain better understanding for MA. There always exist several limitations in research of measurement. Limitations of the study and suggestions for future research are as follows. First, the findings of MASS proved that one MA group, HAX and one AC group, HAC was significantly lessened by CTP. The findings on the brain-imaging data by EEG presented that there was similarity in patterns within the groups, showing that the groups by the MA aspect with higher amplitude were more influenced than the AC aspect on working memory in brain.

Furthermore, we should take care of the group, LAC, that received less help from CTP to solve the issue of supoza which stops learning of mathematics in Korea. It is apparent that functional understanding is built on the understanding of the concepts of variables and of the relation between two variables. Without this prior knowledge, students ought to have difficulty in proceeding to learn functions. Actually, three students among 10 of LAC group, who belonged to LAX, not to HAX, need to be provided with an individualized-learning method as a treatment based on the precise diagnosis. Neither MA nor AC aspects of a student could be better in mathematics without this kind of considered treatments by a teacher because mathematics is built on hierarchical conceptual understanding.

Second, when we planned and conducted the research according to the actual situation of the school site, there was not enough time to apply the program as we

planned. In order to make a shortened version of the program by reducing 10 units to 3 units, we referred to the previous research and had discussions with experts in mathematics education. For further research, it is necessary to expand the complex program not only to cover the material for a longer period of time, but also to involve more students, so that we might increase the validity of the study. Furthermore, for measuring EEG, an additional classroom to build a shield room as a quiet place should be available for administering the ERP to have the experiment successful without the distraction of other noises. For our study, the school we visited had an extra room for us to use.

Third, due to the time constraint mentioned above we carried out only one way titled F-G Model (for students to start with type F and then to proceed to type G), consisting of 40 items (actually 60 items including the arithmetical type). In the following study, the other way called 'G-F Model' should be performed to have the broader range of students' responses so that we could understand more how to lessen students' mathematics anxiety in functional tasks.

Fourth, the students responded faster and more accurately to type G than type F after working with CTP as shown in Table 5. However, EEG results confirmed that the students needed to concentrate more on type G because of the characteristic of type G. Also, when they improved related to anxiety, their brain-imaging data presented some similarity in pattern on functional types. We are aware that it is hard to generalize with a small number of participants, however, if they show some kind of similarity on brain waves in the two types they encounter, it is a signal that they alleviated their math anxiety more or less.

Regarding the functional types, one suggestion could be made about the textbook for students to use every day. As we investigate Korean textbooks to see how they distribute type F and G, few of them include a portion of type G, compared to type F. Our study suggests that some activities for type G be reinforced into the math textbooks along with type F.

Today, the advent of cutting edge-handy devices for measuring brain waves makes it possible to apply them for educational purposes like reducing MA. The importance of mathematics is increasing day by day, but there are many students, who have a sense of discomfort with mathematics. For the sake of national core competence, we should solve the MA issue and develop new methodologies in mathematics education. In this study, the expertise of math educators played a crucial role in developing the type of tasks for EEG because they enabled students to provide the structure of composing mathematical tasks. Therefore, we would like to insist on the necessity of expansion of this kind of studies with other experts such as psychologists and brain-imaging scientists with nationwide support.

Disclosure statement

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Notes on Contributors

Prof. Sang S. Choi-Koh who acquired her degrees of M.Ed and Ph.D at University of Georgia in USA, now works for Dankook University, internationally publishing over 120 articles and over 10 books in the area of mathematics education for the secondary school in the last two decades.

Byeong G. Ryoo is a graduate student who is interested in Math Anxiety measured by EEG. Now he writes his doctoral dissertation about how to reduce students' Math Anxiety in geometric thinking. This also could be an unprecedented work in math education.

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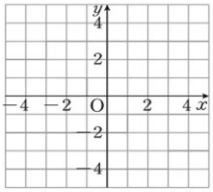
Appendix 1. One sample of a mathematical treatment

Worksheet 2 Class # _____ Name _____

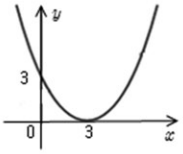
Diagnosis PB | # of hints ( )

Using quadratic function $y = -x^2$, draw the graph of $y = -x^2 + 2$ on the coordinates below.



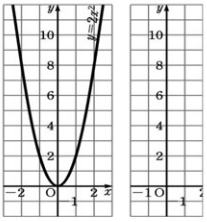
Treatment PB 2 | # of hints ( )

If we translate the graph $y = ax^2$ in a direction parallel by p to the x-axis, then we get the graph below. Find the value of ap, where a and p are consonants.



Basic PB | # of hints ( )

Draw the graph of $y = 2(x-1)^2$ on the coordinates below, using a quadratic equation $y = 2x^2$.

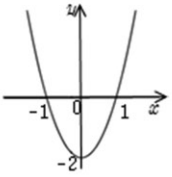


Challenging PB | # of hints ( )

If the graph of function $y = ax^2 + b$ that passes point $(1, -2)$ is being translated in a direction parallel by 2 to the y-axis, then it becomes the function $y = cx^2 + 4$. Find consonants, a, b and c.

Treatment PB 1 | # of hints ( )

Find the equation of the graph below.



Appendix 2. Hints for Worksheet 2

Hint Cards for the Treatment PB 1			
Hint1	If we translate the graph of $y = ax^2$ parallel by q to y-axis, then $y = ax^2$ becomes $y = ax^2 + q$.	Hint 4	It passes points, $(-1,\ 0)$ and $(1,\ 0)$, we insert either one into $y = ax^2 - 2$.
Hint2	Since we translated the graph of $y = ax^2$ parallel by -2 to y-axis, it becomes $y = ax^2 - 2$.	Hint 5	It is $0 = a(-1)^2 - 2$ or $0 = a(1)^2 - 2$, then $a = 2$.
Hint3	The graph of $y = ax^2 - 2$ has the vertex, $(0,\ -2)$ on y-axis.	Hint 6	The answer is $y = 2x^2 - 2$.

Hint Cards for the Treatment PB 2			
Hint1	If we translate the graph of $y = ax^2$ parallel by p to the x-axis, then $y = ax^2$ becomes $y = a(x-p)^2$.	Hint 4	It passes point $(3,\ 0)$ on y-axis, we insert them $(x = 0,\ y = 3)$ into $y = a(x-3)^2$.
Hint2	Since we translated the graph of $y = ax^2$ parallel by 3 to x-axis, it becomes $y = a(x-3)^2$. So $p = 3$.	Hint 5	It is $3 = a(0-3)^2$, then $9a = 3$. So, $a = \frac{1}{3}$.
Hint3	The graph of $y = a(x-3)^2$ has the vertex, $(3,\ 0)$ on x-axis.	Hint 6	The answer is $ap = \frac{1}{3} \times 3 = 1$.